Security Group vs Network Access Control List.

Security Group (SG):

What it is:

A **Security Group** is a **stateful firewall attached to a resource** (like EC2).

What "stateful" really means:

If you allow inbound traffic → **response is automatically allowed**.

You don't need to configure return traffic.

Key points:

- Works at **instance level**
- Only supports **ALLOW rules**
- Default: **deny everything**
- Very simple and safe

Example:

Allow HTTP:

- Inbound: port 80 → ALLOW
  Done. No outbound rule needed.

Network ACL (NACL):

What it is:

A **NACL is a stateless firewall applied at subnet level**.

What "stateless" means:

AWS does NOT track connection state.

👉 You must allow:

- inbound traffic

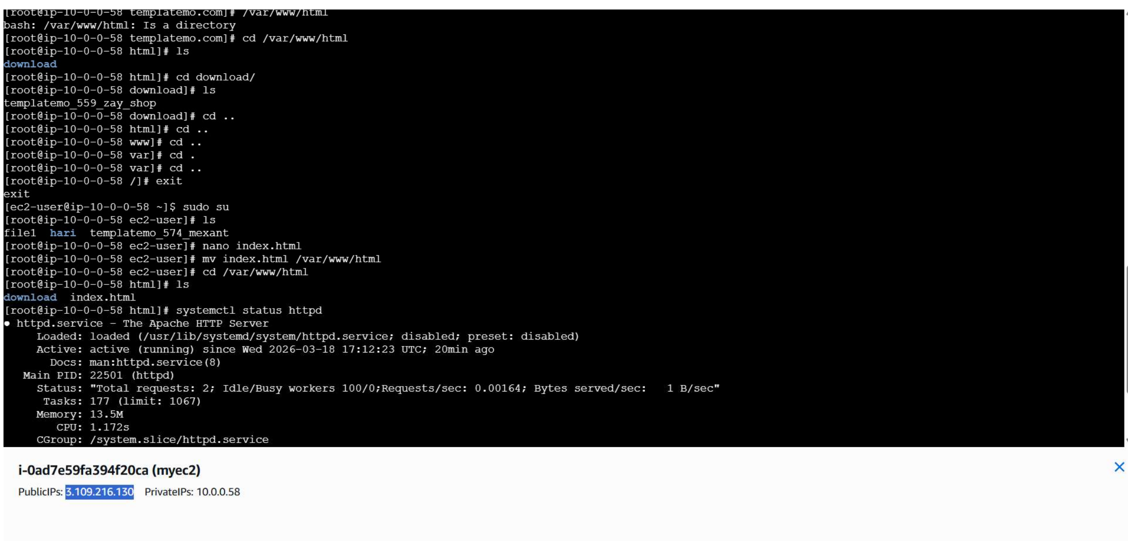
- outbound response separately

Miss one → connection fails.

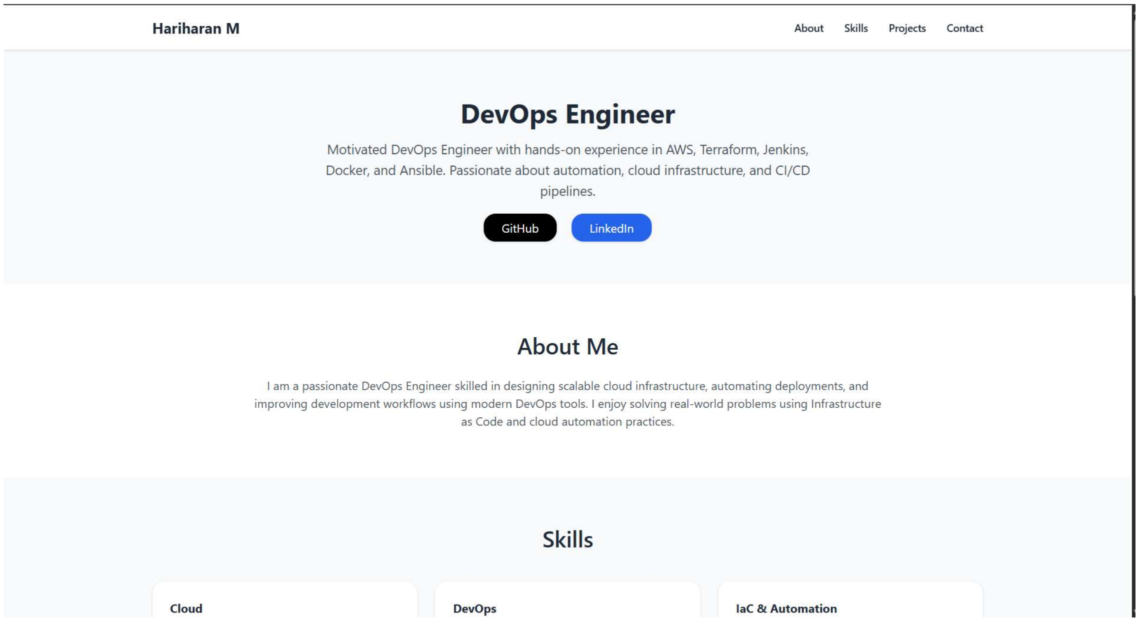
Key points:

- Works at **subnet level**
- Supports **ALLOW + DENY**
- Rules processed in **number order**
- More control, more mistakes

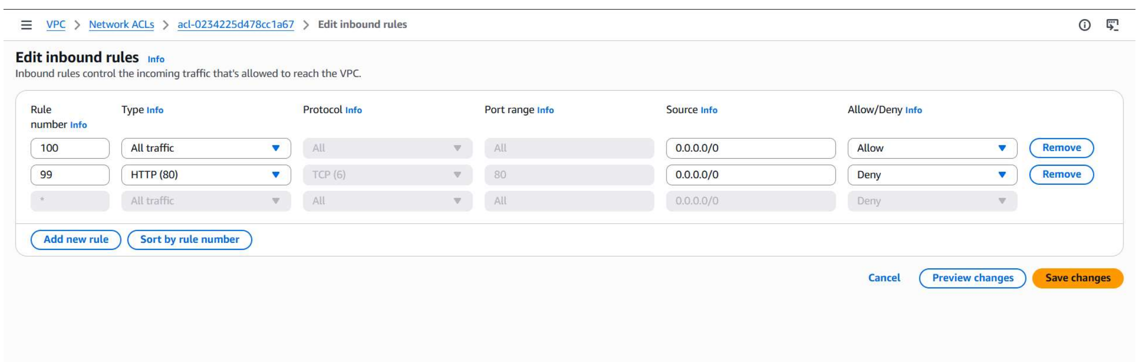
Then I have installed and started the Apache httpd Web Server.



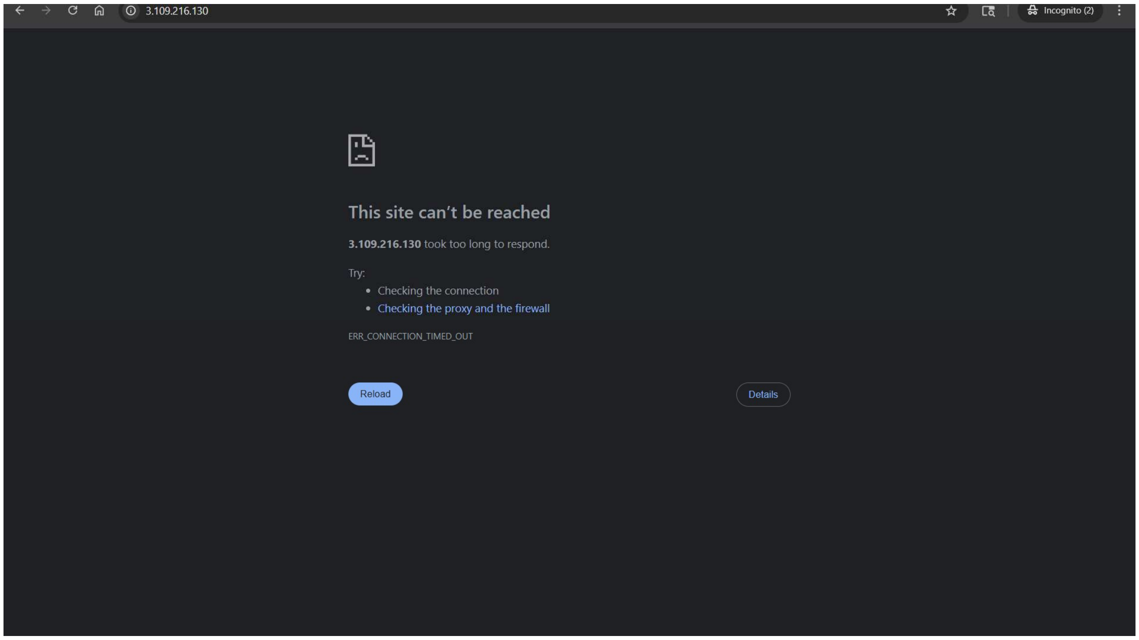
Then I have hosted the http web server my portfolio.



Then using NACL I have denied the http port in the inbound rules.



Then it shows the website can’t be reached.



Elastic IP (EIP)

What it is:

An **Elastic IP** is a **static public IPv4 address** in AWS that you own and can attach to resources.

Why it exists:

Normal public IPs in AWS **change when you stop/start EC2** → useless for real systems.
EIP solves that by giving you a **fixed IP**.

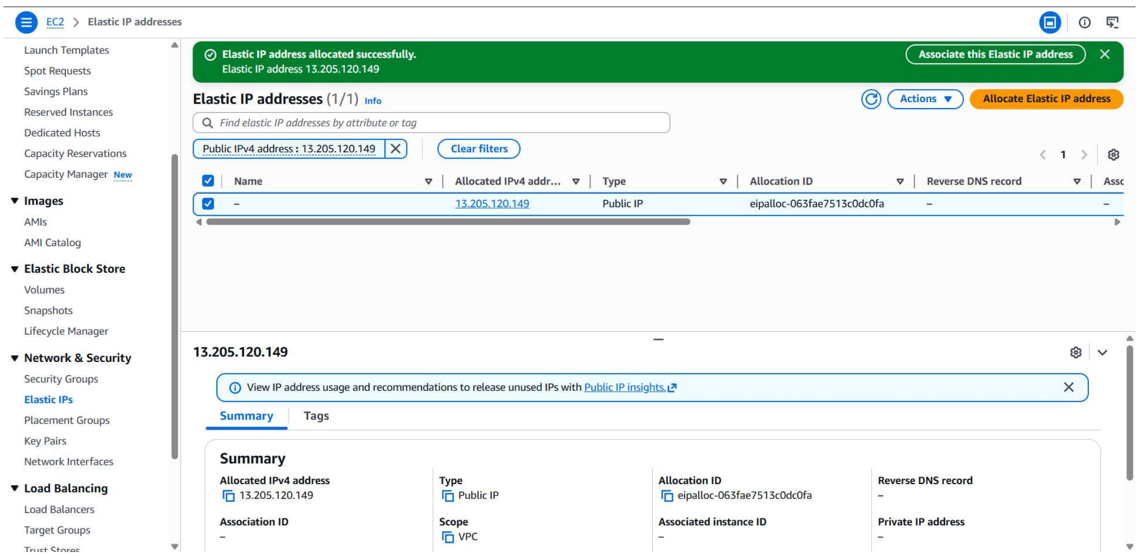
Key points:

- Static (does NOT change)
- You can **attach/detach** from instances
- Works with:
    - EC2
    - NAT Gateway
    - Load Balancer (indirectly)

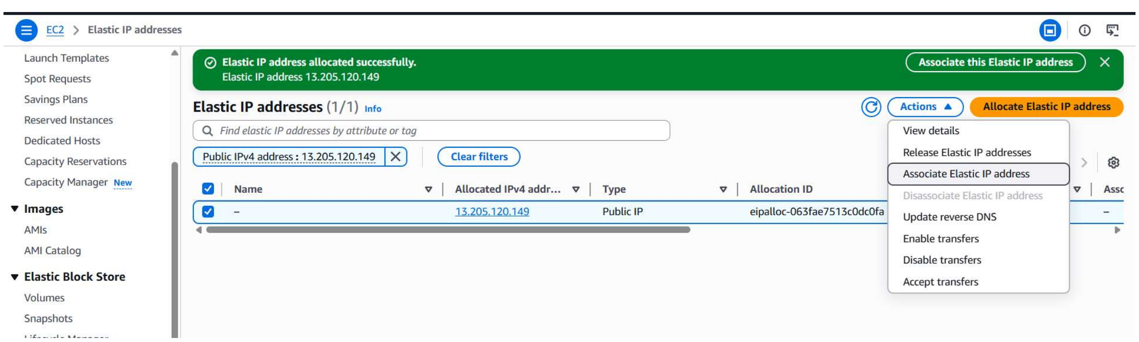
Real use:

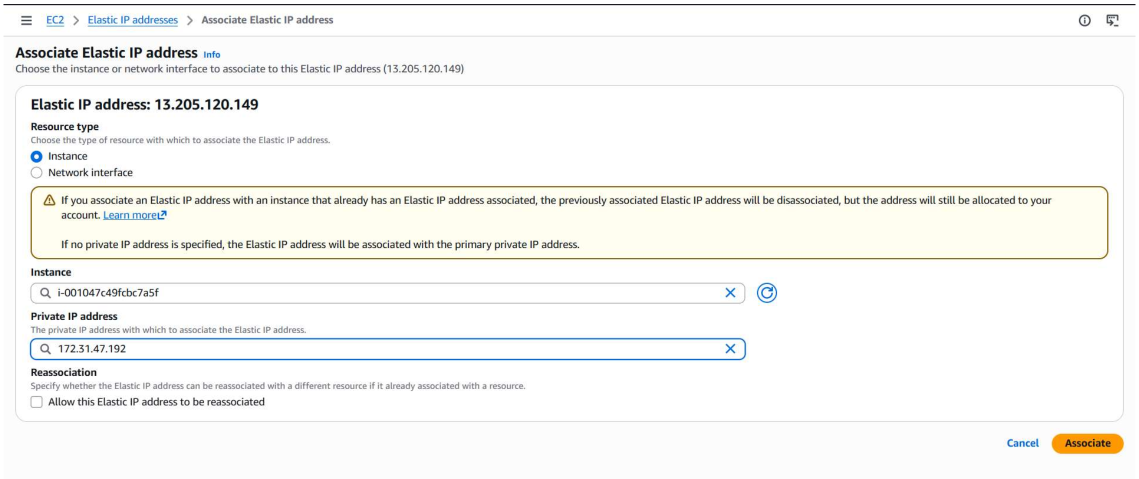
- Hosting a website with a fixed IP
- Whitelisting IP in firewalls
- NAT Gateway outbound internet

Elastic IP Allocation.



After creation we have to associate the Elastic IP.





After association of Elastic IP the public IP is Available after Stopage of Instance.